Software Reuse – Strategic Integration

1. Positioning in the Reuse Landscape

Approach: Software Product Lines (SPL)

Justification: The Najm application serves as a central platform connecting various stakeholders, including insurance companies, government authorities, and policyholders. Adopting an SPL approach is strategic because it allows for a unified infrastructure while catering to the diverse requirements of different insurance providers.

Planning Factors:

- **Development Schedule:** SPL enables rapid deployment of new modules for new insurance partners without reinventing the core engine.
- **Software Criticality:** Since Najm handles legal and financial data, reusing proven, pre-certified core components is essential to ensure high dependability and minimize risks.

2. Product Line Architecture (SPL) Breakdown

Core Components (Generic Functionality):

- Identity & Authentication Engine: Integration with National SSO (Nafath) for secure access.
- Geospatial Reporting Service: The core GPS logic for incident location tracking.
- Unified Notification System: Standardized engine for SMS and Push notifications.

Specialized Components (Configurable Parts):

- Insurer-Specific APIs: Custom integration adapters for various insurance providers (e.g., Tawuniya, Al-Rajhi Takaful) with varying data formats.
- Damage Assessment Modules: Logic branches tailored for different vehicle types (Private vs. Commercial) or accident severities.

3. Validation & "The Ariane 5 Lesson"

The Ariane 5 disaster demonstrated that reusing software in a new context without verifying data constraints leads to failure. For Najm, where GPS coordinates and financial claims are processed, data integrity is paramount.

Validation Plan: We will implement rigorous Regression Testing and Interface Stress Testing. Each reused component will undergo "Boundary Value Analysis" to ensure it handles the scale and data types of the live environment.

Reliable Programming Commandment: "Thou shalt explicitly check all data conversions and input bounds from external sources before execution."

- Implementation: We enforce strict Type-Safety and utilize "Try-Catch" blocks for data-type conversions (e.g., ensuring coordinate precision) to prevent silent overflows.

4. Industry Alignment: Micro-services

Our strategy leverages Micro-services Architecture. By isolating the "Accident Reporting" service from the "Claim Processing" service, we achieve:

- Isolation: A failure in a reused payment gateway component will not crash the incident reporting core.
- Recovery: Individual services can be rebooted or patched independently, ensuring the system remains resilient.

5. From Technical Knowledge to Professional Wisdom

In Assignment 6, our resilience goals focused on maintaining availability and data integrity. In Assignment 7, this Reuse Strategy acts as the primary enabler. By utilizing a Software Product Line, we inherit the stability of battle-tested components. Relying on standardized, reused modules for critical paths reduces the "attack surface" for bugs, ensuring that Najm remains a dependable pillar of the traffic and insurance ecosystem.